

Seng Chon Chang

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EDUCATION

The Pennsylvania State University – Harrisburg

Bachelor of Science in Electrical Engineering

GPA: 4.0/4.0

Relevant Coursework: Calculus and Vector Analysis; Circuits and Devices; Cornerstone Engineering Design; Ordinary and Partial Differential Equations; Programming and Computation I: Fundamentals; Programming and Computation II: Data Structure

Middletown, PA

May 2029 Expected

CAMPUS & COMMUNITY INVOLVEMENT

Chair, Technical Committee, Penn State Harrisburg IEEE Student Branch (8 hrs./wk., 6 mos./yr.)

Middletown, PA

- Mentor/Lead the technical team of 19 to organize 16 1-2 hour-long technical workshops each semester Since Sep 2025
- Organize/Host/Teach 5 technical workshops
 - Upcoming event: Hack the Campus Hackathon, expecting 100+ student participants
 - Past events: Design & 3D-Print a Phone Holder; 3D-Printing Crash Course; Halloween Soldering Workshop; Design & 3D-Print Your Own Key Chain with Fusion 360!; 4-40+ student participants; bit.ly/d-wshop
- Initiated request and received equipment sponsorship (including an A1 3D printer and AMS - Automatic Material System), Bambu Lab, January 2026

Founder/President, Design and Applied Technology club, Instituto Salesiano High School (7 hrs./mo.)

Macao Island, Macao
July 2024 - May 2025

- Promoted STEM interests on campus through technology, programming, design, and innovation
- Recruited 60+ members; established competition, marketing, and operations divisions; hosted weekly officer meetings
- Organized special events: school open-day club showcase (300+ students), Feb 2025; fundraiser booth (700+ public) at the 55th Macao Charity Bazaar, Nov 2024

WORK

Macao One Zero Technology Ltd., facebook.com/10computerMacao

Macao Island, Macao

IT Support Technician & System Solutions Assistant, \$250+/mo. (4 hrs./wk., 2 wks./mo.)

Aug 2022 - Jun 2025

- Completed 180+ laptop/PC repairs; troubleshoot, part replacement, test; customer service & education
- Restored 4 TB+ files from 20+ corrupted hard drives; built 22+ custom PCs
- Co-developed/Maintained an online jewelry shopping website, backend server and frontend UI, July - Aug 2023
- Co-operated weekly service livestream, Macao Baptist Church, on Ubuntu using OBS (scene switching, audio leveling, encoder/stream), Jan 2024

Fuyue Xianju Vegetarian Restaurant, bit.ly/d-vegan

Macao Island, Macao

Waiter, \$8/hr. (40 hrs./wk., 4 wks./mo.)

Summers 2020/2021

- Took orders, served meals, cleared plates and utensils, reset tables, organized dishware, cleaned dining area

EXTRACURRICULAR PROJECTS

Interactive Tools for Teaching Forward and Inverse Kinematics, k.sengchon.com

Dec 2025 - Current

- Develop an online, interactive visual demo and program a physical 3 DoF robotic arm for in-class (ME 456 Introduction to Robotics) demonstration by Dr. Amit Banerjee, Associate Professor, Mechanical Engineering, Penn State Harrisburg

A Bio-Inspired, Flexible, and Retractable Robotic Tail Exoskeleton, isef.net/project/192569

Aug 2024 – May 2025

- Founder/Captain/Team of 2; created a wearable robotic tail that reduces 70% gravitational torque, which negatively correlates with human gait stability
- Drew Inspiration from cats' tails; reviewed literature; designed 83+ iterations in Fusion 360; printed 12+ 3D prototypes; programmed C++ library for BLDC motors; deployed ROS2 on JetsonNano for modularity; integrated nonlinear IK solver for soft robotics; developed a real-time balance algorithm; co-designed evaluation methods

NeRF Technology: An Economic Approach to 3D Scanning, isef.net/project/16240

July 2023 – May 2024

- Member/Team of 3; created a high-quality 3D scanner capable of digitizing translucent museum artifact
- Innovated a pre-processing workflow in Neural Radiance Fields (NeRF); engineered a 3-DoF robotic arm and a stepper-motor rotary table in Fusion 360 (161 iterations); designed & conducted evaluations

A Line-Following, Lightweight, Two-Wheeled Robot, sengchon.com/lbot

Sep 2023 – Nov 2023

- Founder/Captain/Team of 2; designed from scratch a line-following robot with a carbon-fiber chassis for faster movement
- 3D-printed herringbone gearbox with 4.55:1 reduction; implemented power regulation with LM2596 buck (12V-5V); used 2 CD4051B multiplexers to read 15 analog sensors; established PID controller; designed board-to-board protocol; developed a web UI (React + Material UI) with features for connection, driving, PID tuning, calibration

Additional Independent/Individual Maker Projects, sengchon.com

Since 2018

- Brushless DC Motor Driver Based on F.O.C.
- CoreXY-based Drawing Robot
- Rubik's Cube Solving Robot
- Two-legged Servo Robot
- User-Generated Business Review Map Web App

HONORS AND AWARDS

- Honorific Title, Certificates of Merit, team of 2, *A Robotic Tail Exoskeleton*, Government of the Macao Special Administrative Region (MSAR)

Jan 2026
 - 3rd, \$1200, Grand Award, team of 2, *A Robotic Tail Exoskeleton*, Engineering Technology: Statics and Dynamics category, Regeneron ISEF

May 2025
 - Finalist, team of 3, *NeRF Technology: An Economic Approach to 3D Scanning*, Robotics and Intelligent Machines category, Regeneron ISEF

May 2024
 - 2x 1st place, Elite Talent & Innovation Competition, Education and Youth Development Bureau, MSAR

Nov 2024/2023
 - 2nd/3rd places, Selection Competition, Macao Youth for Integrated Robotics Science Activity

May 2022/2021
 - 2nd place, High School Regular, Macao Regional, World Robot Olympiad

July 2021

SKILLS

Technical:

Embedded Systems Design	PCB Design	SMD Soldering
Motor Control (DC: H-bridge; BLDC: F.O.C.)	Power Regulation (DC-DC)	ROS2/micro-ROS Integration
Programming (C++, Python, Java, JavaScript)	3D CAD (Fusion 360; AutoCAD)	PID Control
Prototyping: (FDM/SLA 3D printing, laser cutting)	Forward/Inverse Kinematics Solving	Version Control (Git)
Device Repair/Troubleshooting	Frontend/Backend Development	Computer Vision Implementation

Laboratory:

Oscilloscope/Function Generator Operation	Breadboarding/Perfboard Prototyping	Circuit Troubleshooting
Multimeter (DMM) Measurement	Soldering/Rework (TH/SMD)	Serial Debugging

Language:

English (Full professional)	Cantonese (Native)	Mandarin (Native)
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